

Elucidating ferroptosis mechanisms in heart failure through transcriptomics, single-cell sequencing, and experimental validation

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ABSTRACT

Background: The mechanisms underlying ferroptosis in heart failure (HF) remain incompletely understood.

Methods: This study analyzed the heart failure dataset from the Gene Expression Omnibus to identify differentially expressed ferroptosis-related genes (DFRGs). Key DFRGs were selected using LASSO regression and SVM-RFE machine learning techniques. Their diagnostic accuracy was evaluated via ROC curve analysis. Single-cell sequencing data, heart failure cell, and mouse models were utilized to validate these key DFRGs. Additionally, potential non-coding RNAs targeting these genes were predicted, and analyses for gene set enrichment, immune cell infiltration, and drug targeting were conducted.

Results: A total of 127 DFRGs were identified, with 83 downregulated and 44 upregulated compared to controls. Seven key DFRGs (PTGS2, BECN1, SLC39A14, QSOX1, MLST8, TMSB4X, KDM4A) were identified, showing high diagnostic accuracy (AUC 0.988) in the GSE5406 dataset. GO and KEGG analyses linked these genes to ferroptosis, FoxO signaling, and autophagy pathways. A ceRNA network identified 217 miRNAs and 243 lncRNAs potentially targeting these genes, and 64 drugs were predicted as potential targets. Single-cell sequencing and in vitro experiments revealed differential expression of SLC39A14 and QSOX1, which was further confirmed in vivo.

Conclusion: This study provides novel insights into the role of ferroptosis in heart failure by identifying and validating DFRGs that exhibit differential expression across various cell types. The differential expression patterns of these genes, particularly SLC39A14 and QSOX1, indicate their potential involvement in the pathophysiological mechanisms contributing to HF. These findings offer new insights for the development of targeted therapies for HF.

1. Introduction

Heart failure (HF) as a global health challenge, poses a significant burden on society and individuals due to its rising incidence [1,2]. Despite advancements in HF treatment, its complex pathogenesis remains one of the scientific community's challenges [3]. In recent years, research on ferroptosis, a mechanism of cell death, has gradually gained attention and provided new insights into the pathophysiological process of HF [4,5].

Ferroptosis is an iron-dependent cell death process characterized by

excessive accumulation of intracellular iron ions and increased lipid peroxidation of the cell membrane, leading to membrane rupture and cell death [6]. In the cardiac tissues of HF patients, excessive accumulation of iron ions and increased lipid peroxidation have been observed, which may contribute to cardiac cell death and deterioration of cardiac function [7,7]. Moreover, ferroptosis may exacerbate cardiac injury and functional decline by inducing membrane lipid peroxidation and cytotoxicity in cardiomyocytes [8,9].

Single-cell sequencing technology can detect the gene expression levels of each cell within a population, revealing the roles of different

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cell types in disease progression. This not only aids in the deeper understanding of disease pathogenesis but also helps in discovering new biological mechanisms and therapeutic targets [10–12]. Non-coding RNAs, as important regulators of gene expression, are involved in numerous physiological and pathological processes in the heart by modulating the transcription and translation levels of genes [13,14].

This study aims to utilize machine learning methods and bioinformatics analysis to identify potential key ferroptosis-related genes in HF pathogenesis and to explore the molecular mechanisms and regulatory pathways of ferroptosis, as well as its association with HF. Through this research, we hope to provide new insights into the comprehensive understanding of HF pathogenesis and to offer new directions and possibilities for developing future therapeutic strategies.

2. Materials and methods

2.1. Identification and correlation analysis of key genes in HF

This study searched the Gene Expression Omnibus (GEO) database (<https://www.ncbi.nlm.nih.gov/geo/>, accessed on May 26, 2024) using the term “HF” to find gene expression profile datasets related to HF. The included datasets had to meet the following criteria: (i) contain whole-genome mRNA microarray data; (ii) samples sourced from heart tissue of HF patients and healthy controls; (iii) limited to human species; (iv) a sample size sufficient to support statistical analysis. Ultimately, the largest HF microarray dataset (GSE5406) was included from the GEO database, and the corresponding annotation files were used to annotate the probes. Additionally, the single-cell HF dataset (GSE161470) was downloaded. Using R software (version 4.2.1) and the “limma” package, genes with p -values less than 0.05 and $|\log_2FC| > 0.5$ were defined as differentially expressed genes. The data in the GEO database are publicly available resources, so no additional ethical approval was required for this study (Fig 1).

This study also downloaded 729 ferroptosis-related genes from the FerrDb database (<http://www.zhouan.org/ferrdb/>) and used the “limma” package to filter out differentially expressed ferroptosis-related genes (DFRGs). Additionally, “heatmap” and “corrplot” packages were used to generate heatmaps and correlation analysis plots to display the

results.

2.2. GO and KEGG pathway enrichment analysis of genes

The clusterProfiler package was used to perform Gene Ontology (GO) and Kyoto Encyclopedia of Genes and Genomes (KEGG) functional enrichment analysis. Based on the hypergeometric distribution, differentially expressed mRNAs were subjected to functional enrichment analysis in GO categories and KEGG pathways. The GO analysis primarily includes three components: Molecular Function (MF), Cellular Component (CC), and Biological Process (BP),

2.3. Screening of key DFRGs using LASSO regression model

The “glmnet” package in R software was used to construct the optimal Least Absolute Shrinkage and Selection Operator (LASSO) logistic regression model. Through cross-validation, the tuning parameter (λ) that minimized the mean squared error was selected to establish the best LASSO logistic regression model. The LASSO regression model is particularly suitable for reducing model complexity and effectively identifying key genes closely related to HF by shrinking the coefficients of less important variables to zero for feature selection [15]. The support vector machine-recursive feature elimination (SVM-RFE) algorithm was generated using the R package “e1071”. Initially, the importance of the DFRGs was ranked based on the feature weights of the support vectors. Subsequently, the ranking was conducted according to the feature weights of the support vectors¹⁷. Further, the cross-validation error was estimated by ten-fold cross-validation [16,17].

2.4. Analysis of receiver operating characteristic (ROC)

Genes obtained from both the LASSO and SVM-RFE algorithm models were intersected using a Venn diagram to identify the intersecting genes, referred to as interGenes. The “ROCR” package in R software was utilized to conduct ROC curve analysis, assessing the stability of the two algorithms and their sensitivity in diagnosing HF.

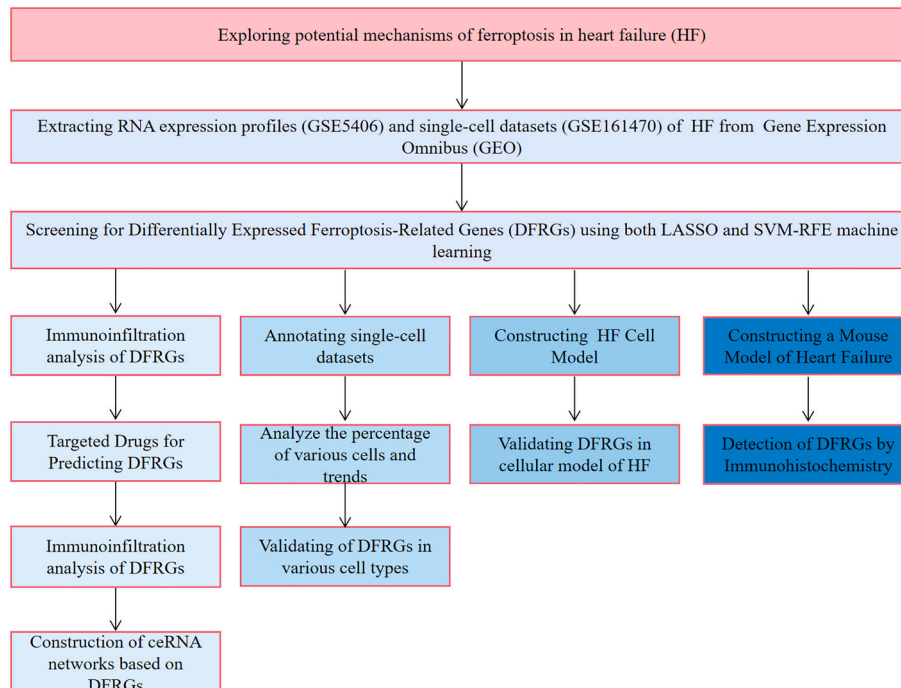


Fig. 1. Flowchart for the study.

2.5. Gene set enrichment analysis (GSEA)

GSEA is typically used to analyze and interpret coordinated pathway-level changes in transcriptome experiments, aiding in the study of underlying mechanisms. In this study, the “clusterProfiler” package in R software was used to perform GSEA, and the “ggplot2” and “enrichplot” packages were utilized for visualization [18]. A gene set was considered significant when it reached an adjusted *p*-value of less than 0.05.

2.6. Correlation between interGenes and infiltrating immune cells

This study employs the CIBERSORT method (<http://cibersort.stanford.edu/>) to analyze mRNA expression profiles for characterizing the cellular composition of the left atrium and normal tissues, encompassing 547 genes across 22 types of immune cells (including 7 types of T cells, naive and memory B cells, plasma cells, NK cells, and myeloid subsets) [19]. Samples with a CIBERSORT *p*-value <0.05 were used for subsequent analyses. Visualization was then conducted using the “reshape2,” “tidyverse,” “ggplot2,” and “vioplot” packages in R software. Specific significant differences were demonstrated when *p* < 0.05.

2.7. Prediction of potential therapeutic drugs using DGIdb

This study utilized the DGIdb database (<https://dgidb.genome.wustl.edu/>) to analyze the relationships among the 7 characteristic ferroptosis-related genes. Based on the prediction results, a drug-gene interaction network was constructed using Cytoscape.

2.8. Prediction of gene-miRNA interactions and lncRNA

The mRNA-miRNA interactions were predicted using TargetScan (<http://www.Targetscan.org/>), miRDB (<http://mirdb.org/>), and miRanda (<http://www.microrna.org/>). Only microRNAs (miRNA) identified by all three prediction tools were considered as key miRNAs [20,21].

2.9. Prediction of gene-miRNA interactions and lncRNA (continued)

The predicted miRNAs were uploaded to StarBase 2.0 (<http://starbase.sysu.edu.cn/index.php>) to predict potential Long non-coding RNAs (lncRNAs) targeting these miRNAs, following very stringent filtering criteria (score > 5). Through this screening process, the lncRNA-miRNA-DFRGs regulatory network was constructed using Cytoscape software [22].

2.10. scRNA-seq quality control and analyses

The “Seurat” R package was used for quality control and analysis of the single-cell HF dataset (GSE161470). The sequencing depth and counts of each cell were initially assessed to remove low-quality single-cell data [23,24]. The top 1500 highly variable genes were extracted, and principal component analysis (PCA) was performed for linear dimensionality reduction. The Louvain clustering method was used to cluster these cells, and marker genes for each cluster were identified. Clustering annotations were performed by integrating published literature, the CellMarker database (<http://xteam.xbio.top/CellMarker/index.jsp>), and annotation results from the “SingleR” R package. The marker genes for each cluster were identified as those genes with the greatest variance in the principal components [25,26].

2.11. Primary reagents and experimental animal

Doxorubicin hydrochloride (ST1285) was purchased from Beyotime Biotechnology. Rodent Diet With 60 Kcal% Fat (D12492) was obtained from Research Diets, USA. N-Nitro-L-arginine methyl ester hydrochloride (N5751) was purchased from Merck, USA. Masson’s trichrome staining kit (G1340) and 4 % cell/tissue fixative were obtained from

Solarbio, Beijing, China. Trizol (AG21102), Evo M-MLV reverse transcription premix kit (AG11728), and SYBR Green Pro Taq HS premix qPCR kit (AG11728) was all sourced from Aikorui Bioengineering Co., Ltd. (Hunan, China). SLC39A14 (Cat No: 26540-1-AP) and QSOX1 (Cat No: 12713-1-AP) antibodies were purchased from Proteintech (Wuhan, China).

Twenty 8 weeks SPF-grade C57BL/6 male mice, weighing (19 ± 3) g, were purchased from Beijing Vital River Laboratory Animal Technology Co., Ltd. Animal experiments were conducted in accordance with the relevant regulations of the Experimental Animal Ethics Committee of Jiangsu Hanjing Biotechnology Co., Ltd. (Ethics Approval No: HJSW-24030401).

2.12. Construction of HF cell model

AC16 cells were placed in a 37 °C, 5 % CO₂ incubator. Log-phase AC16 cells were divided into the Control group and the DOX group (treated with 4 μmol/L Dox for 24 h) to establish the HF cell model.

2.13. Construction of the HFpEF animal model

Twenty 8 weeks male C57BL/6 N mice were acclimated for one week and then randomly divided into two groups based on their weight: the control group (Chow) and the model group (HFpEF), with 10 mice in each group. The control group was fed a standard diet, while the model group was fed a 60 kcal% high-fat diet and water containing N-nitro-L-arginine methyl ester hydrochloride (0.5 g/L). Five weeks after starting the HFpEF diet, cardiac ultrasound examinations were conducted, primarily measuring left ventricular diastolic function (E/A), left ventricular fractional shortening (FS%), and left ventricular ejection fraction (LVEF%) [27].

2.14. Ultrasound detection of mouse heart

The ultrasound detection was performed using the Vevo 3100LT system and MX250 probe. Mice were anesthetized (maintained at 2.0 L/min oxygen flow rate, 1.5 %–2.0 % isoflurane), kept at 38.0 °C, and maintained a heart rate of 400–500 beats per minute. Long-axis images of the left ventricle were obtained in the parasternal long-axis view. Ultrasound analysis was conducted using Vevo LAB, recording at least 5 cycles of systolic and diastolic function on average.

2.15. Tail artery blood pressure monitoring in mouse

The blood pressure of experimental mice was monitored using the non-invasive CODA tail artery blood pressure monitoring system (Sof-ton, BP-2010 A). Blood pressure changes were observed, and values with volumes greater than 20 and standard images were selected as valid blood pressure readings. Blood pressure was measured continuously for three days at fixed times, and the stable average values from days 3 and 4 were used as the mice’s blood pressure.

2.16. RNA extraction, reverse transcription, and real-time quantitative PCR (RT-qPCR)

Total RNA was extracted using the TRizol method. The purity and concentration of the RNA were determined using a NanoDrop ND-1000 UV spectrophotometer. After reverse transcription of the extracted RNA using a reverse transcription kit, RT-qPCR was performed using the SYBR Green method. The reaction system consisted of 20 μL, with initial denaturation at 95 °C for 1 min, followed by 40 cycles of 95 °C for 15 s, 60 °C for 20 s, and 72 °C for 30 s. β-actin was used as the internal reference gene. Primer sequences are provided in Table 1.

Table 1
Primer sequences.

Gene name	Preverseimereverse sequence (5' to 3')
hsa-BECN1-forward	ATCTCGAGAAGGTCCAGGCT
hsa-BECN1-reverse	CTGTCCACTGTGCCAGATGT
hsa-MLST8-forward	CAGGTGAATGCCTTGGAGGT
hsa-MLST8-reverse	ACATGCGGATGTGCTGGTAA
hsa-KDM4A-forward	GAGCCCCTGTGGCAAAGTAT
hsa-KDM4A-reverse	TGGTGTGAAGCCCTCCAATC
hsa-PTGS2-forward	AGGCTTCCATTGACCAGAGC
hsa-PTGS2-reverse	TCCACAGCATCGATGTCACC
hsa-QSOX1-forward	CTTTGGAGGGTGCCAGGTTAG
hsa-QSOX1-reverse	GAGGCTGGTCTCTGTGCATT
hsa-SLC39A14-forward	TCCGAGCGCCAGGTTTATTC
hsa-SLC39A14-reverse	CCATAAGCCAAGCAGGGTCA
hsa-TMSB4X-forward	GAGAAGCAAGCAGGCGAATC
hsa-TMSB4X-reverse	AGGGGACAGCAGTCATTTA
hsa-GAPDH-forward	CCGTTGAATTGCGGTGA
hsa-GAPDH-reverse	TGATGACCCCTTTGGCTCCC

2.17. Masson staining and immunohistochemistry

Heart tissues from each group of mice were fixed in 4 % paraformaldehyde for 24 h, dehydrated, embedded in paraffin, and sectioned. Masson staining and immunohistochemistry were performed according to standard procedures. The sections were digitally scanned using a Panoramic 250FLASH microscope slide scanner, and quantification was performed using ImageJ-win64 software.

2.18. Statistical analysis

Data processing and analysis were performed using GraphPad Prism 9. Statistical data are presented as the mean $\pm$ standard error from three independent experiments. An unpaired *t*-test was used for statistical analysis between two groups, with $P < 0.05$ considered statistically significant.

3. Result

3.1. A total of 127 DFRGs were obtained

After merging and organizing the HF dataset (GSE5406), a total of 356 FRGs were identified (Supplementary Table 1). Among them, 127 FRGs exhibited differential expression (Supplementary Table 2). Compared to the control group, the expression of 83 genes, including CHAC1, was downregulated, while the expression of 44 genes, including PTGS2, was upregulated (Fig. 2A-B). Correlation analysis of these 127 DFRGs yielded a correlation matrix plot. As shown in Fig. 2C, each cell in the correlation matrix represents the correlation between the expression of two DFRGs. Red and blue indicate positive and negative correlations, respectively. The correlation coefficients are displayed in the lower left of the matrix, and their significance is shown in the upper right.

3.2. Functions and pathways of DFRGs

To further explore the main molecular functions and signaling pathways of these 127 DFRGs, GO and KEGG enrichment analyses were conducted. GO enrichment analysis indicated that DFRGs were enriched in several GO terms, including cellular response to chemical stress and response to oxidative stress (Fig. 2D). KEGG enrichment analysis showed that DFRGs were primarily enriched in pathways such as Ferroptosis, FoxO signaling pathway, and Autophagy (Fig. 2E).

3.3. Machine learning were used to further identify to 7 key DFRGs

Two machine learning methods, LASSO regression and SVM-RFE algorithm, were used to identify seven key DFRGs: PTGS2, BECN1,

SLC39A14, QSOX1, MLST8, TMSB4X, and KDM4 A (Fig. 2F-H). Subsequent ROC curve analysis of these seven DFRGs in the GSE5406 dataset demonstrated an AUC of 0.988 (Fig. 3A). To ensure the robustness of these findings, we further validated the results using two additional heart failure datasets (GSE76701 and GSE8331), both of which yielded an AUC of 1, confirming the high diagnostic accuracy and reliability of the seven DFRGs (Fig. 3B-C). Additionally, GSEA was performed based on the data for these seven key DFRGs (Fig. 4D-J).

3.4. Correlation analysis of DFRGs with immune cell infiltration

No significant immune cell infiltration was observed between the control and HF groups (Fig. 4A). A correlation heatmap was generated based on the expression levels of the 7 key DFRGs and the infiltration levels of 22 types of immune cells (Fig. 5B). The expression level of TMSB4X was positively correlated with M2 macrophages and negatively correlated with regulatory T cells and other immune cells. The expression level of SLC39A14 was positively correlated with CD8 T cells and neutrophils, and negatively correlated with NK cells and monocytes. QSOX1 was negatively correlated only with regulatory T cells and other immune cells (all $P < 0.05$).

3.5. The ceRNA network of DFRGs was constructed

The combination of three prediction data pathways, TargetScan, miRDB and miRanda, resulted in the prediction of a total of 217 miRNAs that could potentially regulate the seven key DFRGs. In addition, 243 potential lncRNAs targeting these miRNAs were identified, and a ceRNA network map was constructed (Fig. 4C).

3.6. Predictions were obtained for drugs that might target the DFRGs

Potential target drugs associated with key DFRGs were predicted using the DGIdb database. These drugs may target HF by modulating the key DFRGs. A total of 64 targeting drugs were predicted, including 30 potential drugs associated with KDM4A, 3 drugs associated with MLST8, 30 drugs associated with PTGS2, and 1 drug associated with SLC39A14 (Fig. 4D).

3.7. The key DFRGs were predicted and analyzed by single cell sequencing

The cell types in the HF dataset were divided into 25 clusters with tSNE and UMAP methods. The “FindAllMarkers” function was employed to identify DEGs in each cluster ($\log_{2}FC = 0.25$). Through the use of 10 subtype-specific DEGs for each cell subtype, 10 cell types were identified: B cells, Cardiomyocytes, Dendritic cells, Endothelial cells, Fibroblasts, Macrophages, Mast cells, Neutrophils, NK cells, and T cells (Fig. 5A). Additionally, the proportions of these cell types in the HF and control groups were evaluated. Compared to normal tissue, the percentages of cardiomyocytes, neutrophils, and B cells were reduced in HF tissues, while the percentages of macrophages, endothelial cells, dendritic cells, NK cells, and T cells were increased (Fig. 5B).

Combining the previously identified key DFRGs, TMSB4X was mainly expressed in T cells, NK cells, and macrophages. MLST8 was predominantly expressed in mast cells and fibroblasts. QSOX1 was mainly expressed in T cells, mast cells, and fibroblasts. SLC39A14 was primarily expressed in neutrophils, mast cells, and cardiomyocytes (Fig. 5C). Additionally, the expression levels of SLC39A14 in cardiomyocytes and neutrophils in the control group were lower than those in the HF group. The expression levels of QSOX1 in neutrophils, mast cells, and T cells in the control group were higher than those in the HF group (Fig. 5D).

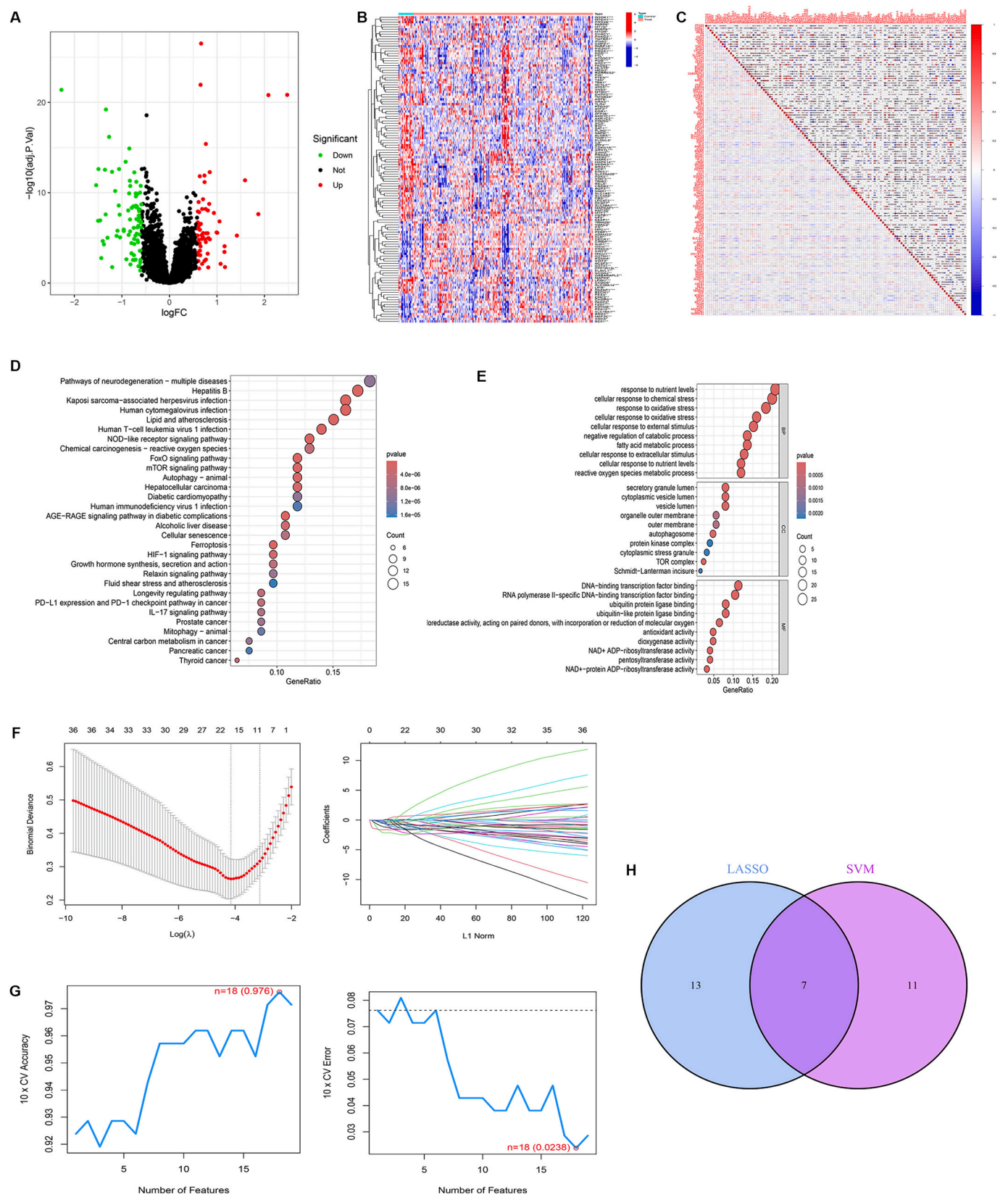


Fig. 2. Screening to DFRGs from HF dataset. (A). Volcano map of GSE5406. (B). The heatmap of 127 DFRGs in GSE5406. (C). Correlation between the expression of 127 DFRGs. (D-E). The main GO terms and KEGG pathways enriched by differentially expressed mRNAs. (F). The performance in of ten-time cross-verification for tuning parameter in selection LASSO. Each coefficient curve in the upper picture represents a single gene. The solid vertical lines in another picture represent the partial likelihood deviance SE, and the number of genes ($n = 10$) corresponding to the lowest point of the curve is the most suitable for LASSO. (G). Accuracy graph and cross validation error graph of DFRGs screened with the SVM-RFE algorithm. (H). Venn diagram of the intersection of DFRGs (PTGS2, BECN1, SLC39A14, QSOX1, MLST8, TMSB4X and KDM4A) screened with the two machine learning methods.

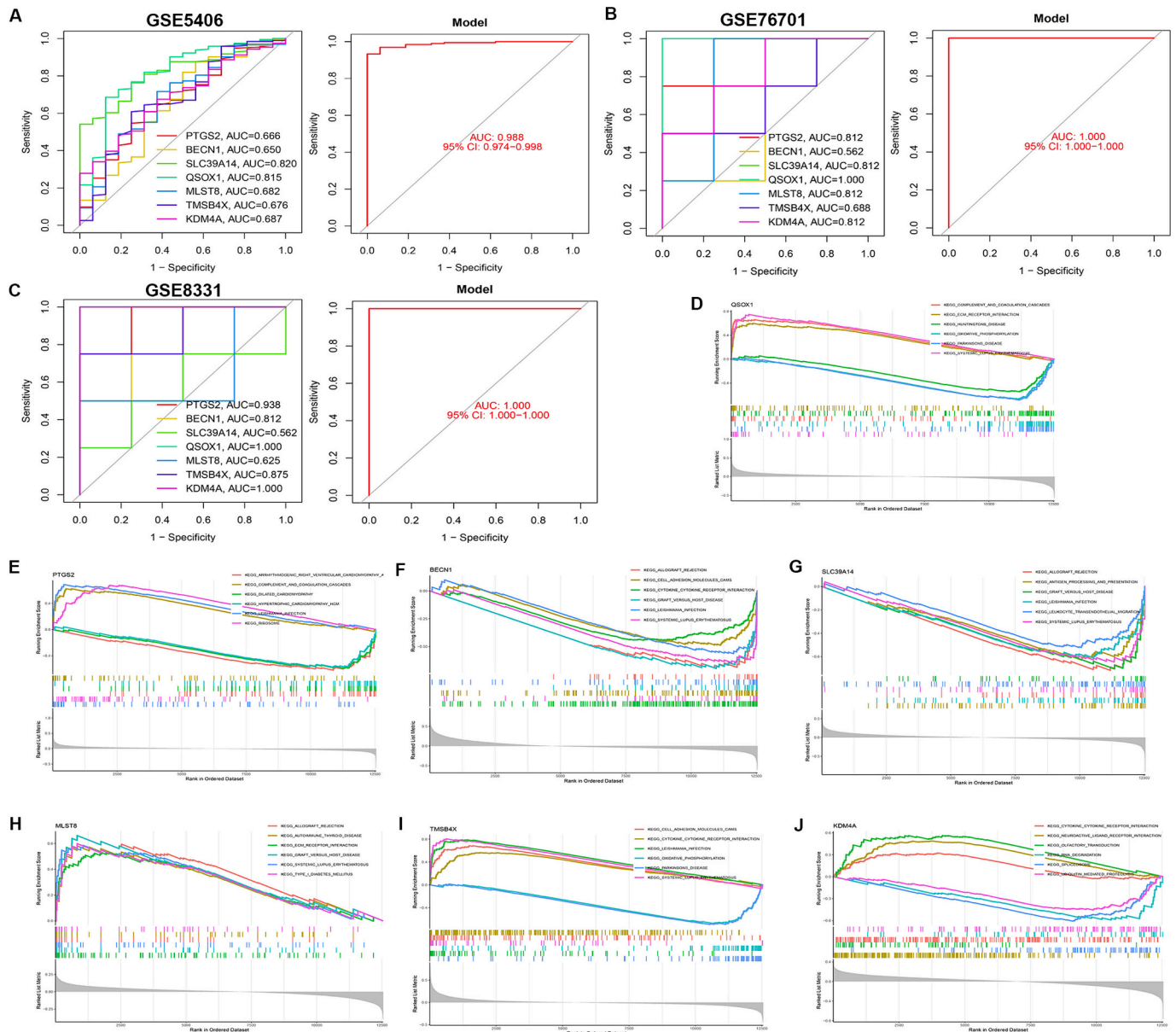


Fig. 3. ROC curves and GSEA results for DFRGs. (A). ROC curve analysis of these seven DFRGs in the GSE5406 dataset demonstrated an AUC of 0.988. (B–C). Validation of DFRGs was supplemented using ROC curve analysis in two heart failure datasets (GSE76701 and GSE8331). (D–J). The 7 DFRGs were enriched separately with GSEA.

3.8. Further validation in the HF model resulted in QSOX1 and SLC39A14

After successfully constructing the HF cell model, RT-qPCR was used to detect the expression of key DFRGs. Analysis revealed that BECN1, TMSB4X, and KDM4A did not exhibit expression changes between the two groups, while MLST8, PTGS2, QSOX1, and SLC39A14 showed differential expression between the two groups (Fig. 5E).

3.9. The HFpEF mouse model was successfully constructed

Results showed that by the third week of modeling, the body weight of the control group mice (24.62 ± 0.83 g) began to show a statistical difference compared to the HFpEF group mice (26.26 ± 1.21 g) ($P < 0.05$). By the fifth week, the body weight of HFpEF group mice (35.37 ± 2.38 g) was significantly higher than that of the control group mice (28.92 ± 0.75 g), with the HFpEF group mouse showing a significant

increase in weight over time (Fig. 6A). The systolic and diastolic blood pressures of HFpEF group mice (SBP: 68.13 ± 2.78 ; DBP: 92.50 ± 3.18) were significantly higher than those of the control group (SBP: 87.83 ± 3.06 ; DBP: 121.7 ± 3.02), with statistical significance (Fig. 6B). At the fifth week of modeling, the left ventricular ejection fraction (LVEF) and left ventricular fractional shortening (LVFS) in the HFpEF group (57.52 ± 2.13 ; 31.47 ± 1.76) showed no significant statistical difference compared to the control group (56.22 ± 1.45 ; 29.01 ± 1.07) (Fig. 6C). The E/A ratio in the HFpEF group (1.465 ± 0.06) was significantly higher than that in the control group (2.12 ± 0.05) (Fig. 6D). These data indicate that the HFpEF diet had no significant effect on systolic function but severely impaired diastolic function in mice, demonstrating the successful construction of the HFpEF mouse model.

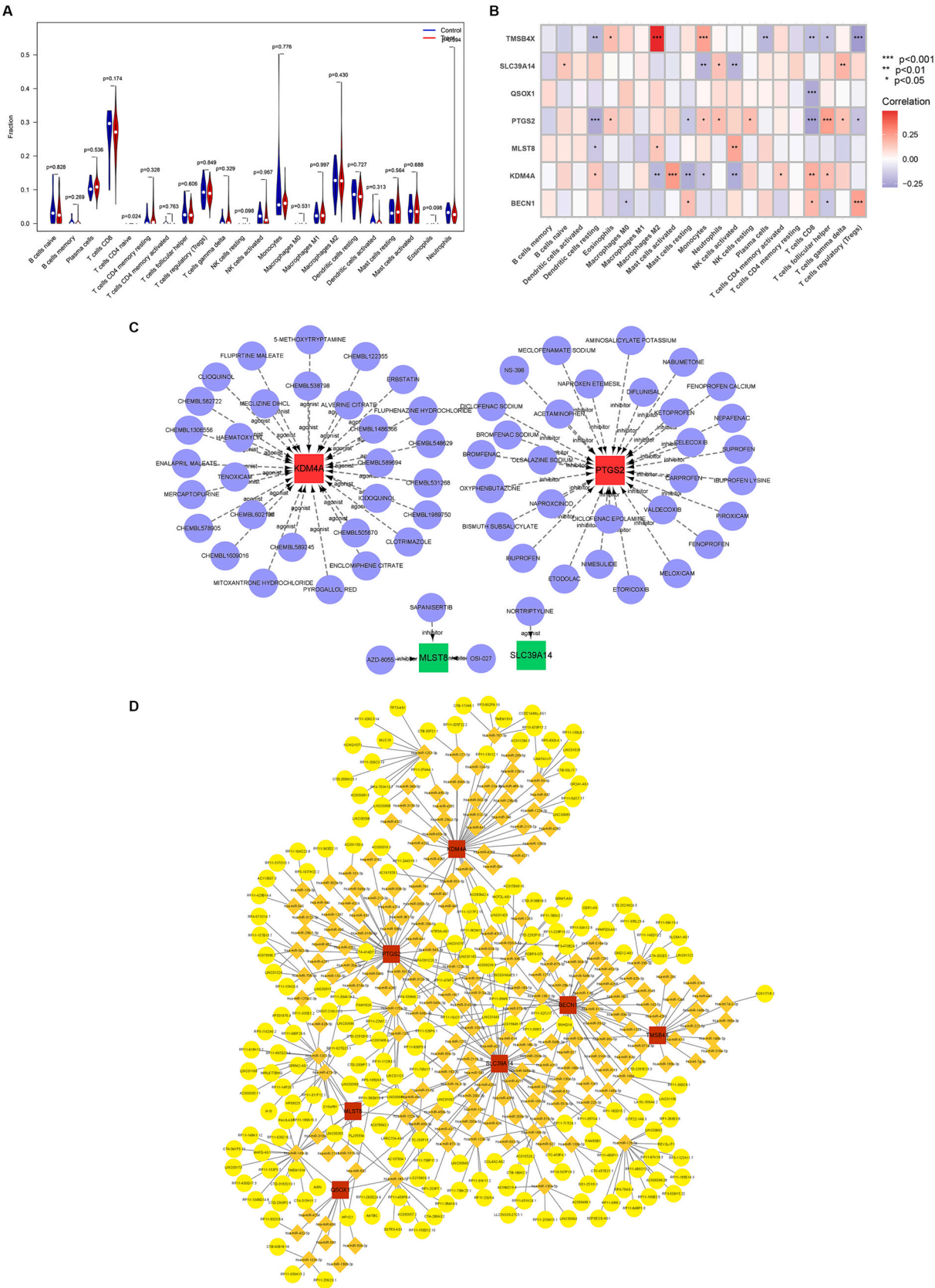


Fig. 4. Immune cell infiltration analysis, drug target prediction and ceRNA network establishment. (A). Differences in the infiltration of 22 types of immune cells in HF. (B). Heat map of the correlation between the expression levels of the 7 DFRGs and the infiltrating numbers of 22 types of immune cells in each group. Red squares: positive correlation; purple squares: negative correlation. Darker color indicates stronger correlation. Not significant (ns) $P > 0.05$, $*P < 0.05$, $**P < 0.001$, $***P < 0.001$. (C). Drug target prediction for 7 key DFRGs using DGIdb database. (D). The ceRNA network of miRNAs and lncRNAs was established based on 7 DFRGs. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

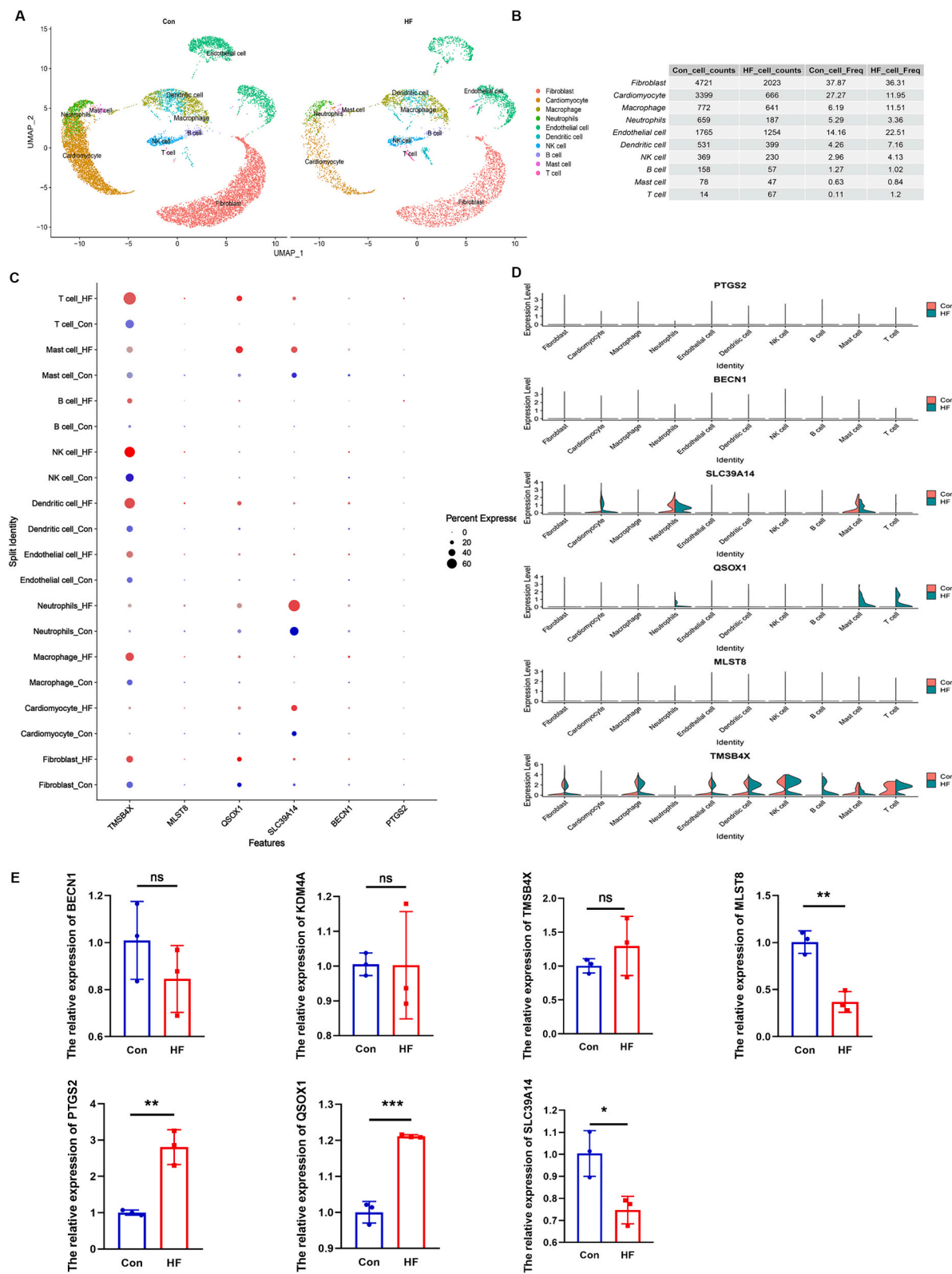


Fig. 5. Expression of 7 key DFRGs in the single-cell sequencing dataset. (A). 10 cell types were identified, including B cell, Cardiomyocyte, Dendritic cell, Endothelial cell, Fibroblast, Macrophage, Mast cell, Neutrophils, NK cell, T cell. (B). Compared with normal tissue, cardiomyocytes, neutrophils, and B cells accounted for a decreased percentage of total cells and macrophages, endothelial cells dendritic cells, NK cells, and T cells accounted for an increased percentage of total cells in heart failure tissue. (C—D). Expression of 7 DFRGs in 10 cell types. MSB4X was mainly expressed in immune cells such as T cells, NK cells, and macrophages, MLST8 was mainly expressed in mast cells and fibroblasts, QSOX1 was mainly expressed in T cells, mast cells, and fibroblasts, and SLC39A14 was mainly expressed in neutrophils, mast cells, and cardiomyocytes. (E). BECN1, TMSB4X, and KDM4A had no altered expression in control and HF cell models, whereas the expression of MLST8, PTGS2 and QSOX1 and SLC39A14 differed between the two groups.

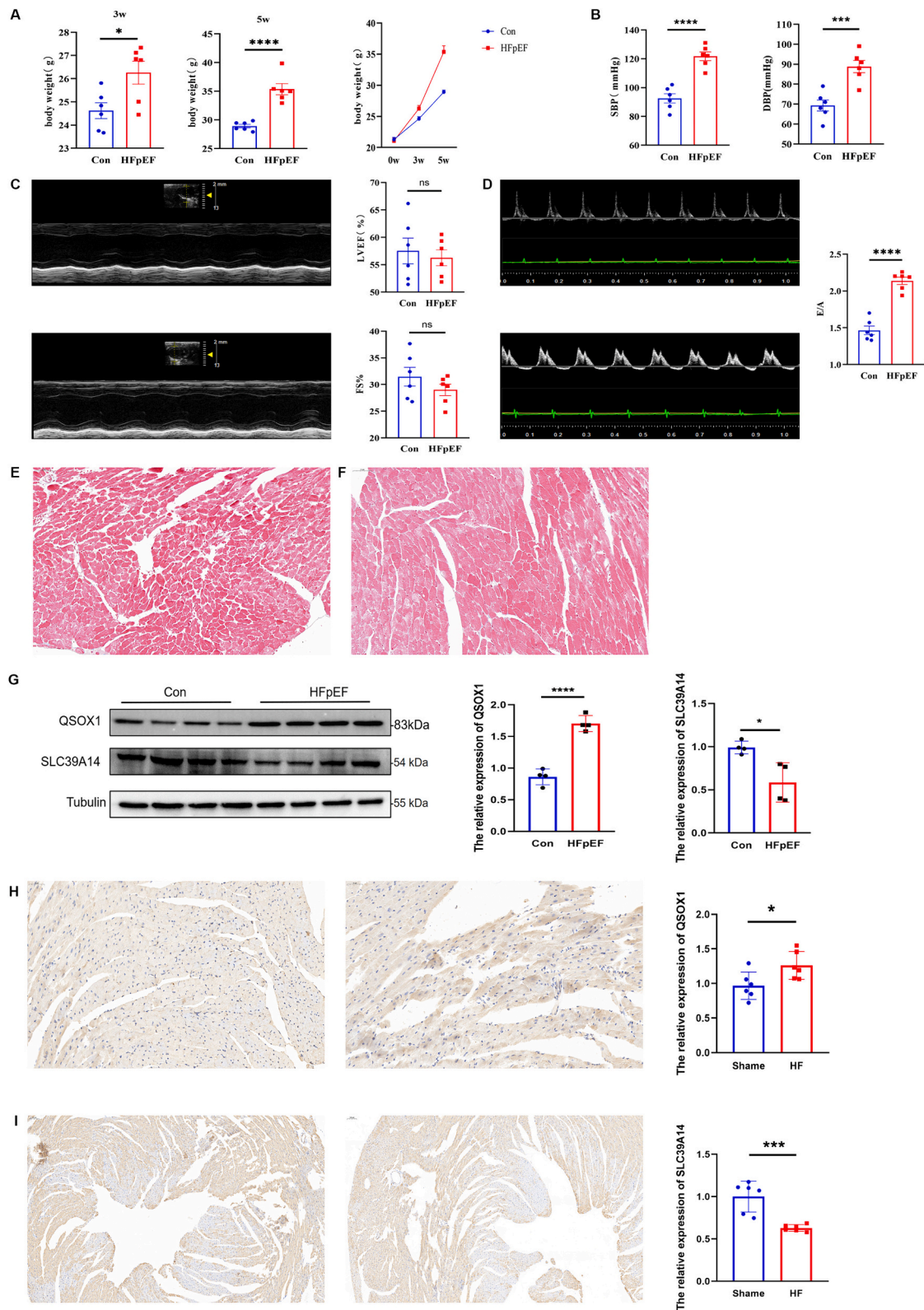


Fig. 6. Differential expression of QSOX1 and SLC39A14 was verified in the vivo. (A). Body weight levels of mice in each group during HFpEF modeling. (B). Changes in blood pressure in each group. (C). There were no statistically significant differences in left ventricular ejection fraction and left ventricular shortening fraction in the HFpEF group of mice compared to the control group. (D). The E/A ratio was significantly higher in the HFpEF group than in the control group. (E-F). MASSON staining of the HFpEF and control groups. (G). Western blot results showed that the protein expression of QSOX1 was higher than that of the control group, and the protein expression of SLC39A14 was lower than that of the control group. (H). The expression of QSOX1 was found to be higher in the HFpEF group than in the control group with the use of immunohistochemical staining. (I). The expression of SLC39A14 was found to be higher in the HFpEF group than in the control group with the use of immunohistochemical staining.

3.10. differential expression of QSOX1 and SLC39A14 was validated in the HF mouse model

As shown in Figs. 6E-F, the myocardial fibers in the control group mice were neatly arranged with minimal collagen fiber deposition. In contrast, the HFpEF mice showed disorganized myocardial fibers, visible fibrosis in the stroma, and enlarged cardiomyocytes compared to the control group. Immunohistochemistry results indicated that QSOX1 expression was higher in the heart tissue of HFpEF mice compared to the control group, while SLC39A14 expression was lower in the HFpEF group compared to the control group (Figs. 6G-I).

4. Discussion

In this study, we employed machine learning methods to identify seven key DFRGs from the HF dataset in the GEO database: PTGS2, BECN1, SLC39A14, QSOX1, MLST8, TMSB4X, and KDM4A. These genes provide crucial insights into the role of ferroptosis in the pathogenesis of HF. We further investigated the expression of these genes across different cell types using single-cell sequencing analysis and validated the potential of SLC39A14 and QSOX1 as diagnostic biomarkers for HF that are closely related to ferroptosis through in vitro and in vivo experiments.

Initially, we screened 127 DFRGs from the HF dataset and further refined this to seven key DFRGs using LASSO regression and SVM machine learning methods. ROC curve analysis across multiple HF datasets suggested that these seven DFRGs may serve as critical biomarkers for HF diagnosis.

Subsequently, we conducted a thorough analysis of the biological functions of these seven DFRGs, revealing their potential applications in early diagnosis, prognosis prediction, and disease progression monitoring of HF. Changes in PTGS2 and SLC39A14 expression may reflect early inflammatory responses and disruptions in metal ion homeostasis [28,29]. BECN1, a key molecule in autophagy regulation, may indicate the presence of myocardial autophagy [30]. QSOX1 and TMSB4X are associated with myocardial fibrosis and repair, respectively, suggesting early damage to cardiac structure and function [31]. MLST8, a component of the mTOR signaling pathway, is involved in cardiac metabolic regulation, while KDM4A, a histone demethylase, influences gene expression and cell fate [32]. These findings, combined with the latest research, offer new avenues for personalized HF therapy and could aid in early detection, long-term prognosis assessment, and disease monitoring. Enrichment analysis of the DFRGs also highlighted potential mechanisms of ferroptosis in HF. KEGG and GO enrichment analyses revealed that DFRGs were predominantly enriched in pathways related to ferroptosis, FoxO signaling, and autophagy, all of which play critical roles in cellular stress responses and maintaining cellular stability [33].

To further validate the seven key DFRGs identified through the two machine learning methods, we assessed their expression levels in cell experiments. We integrated the results from the cell experiments with single-cell analysis, taking into account the biological functions of the cells, and preliminarily selected QSOX1 and SLC39A14 for further investigation. Finally, we validated the differential expression of QSOX1 and SLC39A14 in animal experiments, confirming their roles as key ferroptosis-related biomarkers for diagnosing HF. Further investigation revealed that QSOX1, a thiol oxidase involved in disulfide bond formation and extracellular matrix remodeling, is upregulated in HF cells. Studies have shown that elevated QSOX1 expression in HF cells may be related to myocardial fibrosis and oxidative stress, marking structural changes in the myocardium, and may correlate with HF severity and prognosis in clinical samples [34,35]. Moreover, since QSOX1 is primarily expressed in immune cells such as neutrophils and T cells, we hypothesize that QSOX1 may play a role in extracellular matrix formation and repair [36]. Interestingly, we found that SLC39A14 expression was decreased in HF cell and mouse models, whereas single-cell analysis showed increased expression in myocardial cells and neutrophils in the

HF group. We speculate that changes in the cellular microenvironment due to HF may lead to different expression patterns of SLC39A14 in various cell types, resulting in discrepancies between experimental models and single-cell analysis. Additionally, transcriptome analysis reflects overall changes, while single-cell analysis provides insights into changes within specific cell types. SLC39A14 is a zinc transporter involved in various biochemical reactions, and zinc ion dysregulation can affect iron oxidation and inflammatory responses [37,38]. We hypothesize that reduced SLC39A14 expression in HF cells may alter intracellular zinc levels, subsequently affecting zinc-related enzyme activity and signaling pathways, potentially contributing to HF pathogenesis. Other studies have reported that zinc plays a vital role in antioxidant defense, and decreased SLC39A14 expression may influence oxidative stress by modulating intracellular zinc levels, thereby contributing to HF progression.

Furthermore, we analyzed the correlation between DFRG expression and immune cell infiltration levels. SLC39A14 was positively correlated with CD8⁺ T cells and neutrophils and negatively correlated with NK cells and monocytes. QSOX1 was associated with T cell infiltration levels, consistent with the single-cell analysis results. Additionally, we constructed a ceRNA network targeting DFRGs by predicting miRNA and lncRNA interactions. Using the DGIdb database, we identified 64 drugs that may target the seven DFRGs, including nortriptyline, which may target SLC39A14, although no drugs were predicted to target QSOX1. These findings provide new perspectives for exploring the molecular mechanisms of HF and identifying potential therapeutic targets.

This study still has many shortcomings. First, the study relied on cross-sectional data from the GEO dataset and did not consider longitudinal changes in gene expression. Second, single-cell RNA sequencing validation was limited to the dataset analysis and the results obtained from the analysis differed from those obtained from common transcriptome and experimental validation with no additional single-cell experimental validation. Immune cell infiltration analyses showed no significant differences, suggesting possible issues of sensitivity or dataset limitations. In addition, there was a lack of analysis and validation of key immune cells.

5. Conclusion

This study revealed for the first time the mechanism of iron death in HF by bioinformatics analysis and identified seven key DFRGs. The differential expression of these genes in different cell types highlights their important role in disease progression. Although the study provides new therapeutic directions, there are still some shortcomings, and future studies are needed to further validate these findings to provide more reliable therapeutic strategies.

Ethics approval and consent to participate

Animal experiments were conducted in accordance with the relevant regulations of the Experimental Animal Ethics Committee of Jiangsu Hanjing Biotechnology Co., Ltd. (Ethics Approval No: HJSW-24030401).

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CRediT authorship contribution statement

Kaiyuan Li: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Formal analysis, Data curation, Conceptualization. **Peng Liu:** Writing – original draft, Data curation, Conceptualization. **Lingyu Han:** Writing – original draft, Visualization,

Data curation. **Jing Tian**: Writing – review & editing, Data curation. **Zhipeng Zheng**: Writing – review & editing, Validation. **Min Sha**: Writing – review & editing. **Jun Ye**: Writing – review & editing, Formal analysis, Data curation. **Li Zhu**: Writing – review & editing, Validation, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.cellsig.2024.111416>.

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